**National University of Lesotho**
*Department of Physics and Electronics*

*EE3304 - Electrical Circuit Analysis*
*Semester #1*

**Lab 01: Thévenin/Norton equivalents & maximum power transfer**

---

## 1 Lab Brief and Purpose

This lab explore the concepts of Thevenin/Norton circuit equivalents and maximum power transfer. You should complete the pre-lab exercises before coming to the lab.

## 2 Pre-Lab Exercises

1. For the network shown in Figure 1
   (a) Find the Thevenin and Norton equivalents of the network as seen looking into the terminals *ab*. [$v_{\mathrm{IN}} = 5\mathrm{V}$, $R_1 = 50\ \Omega$, $R_2 = 1.5\ \mathrm{k}\Omega$, $R_3 = 2.2\ \mathrm{k}\Omega$, and $R_4 = 3.3\ \mathrm{k}\Omega$]
   (b) Use Matlab to graph the $i - v$ relation of the network port using the Thevenin/ Norton parameters.
2. What value of the resistance will you connect at the terminals *ab* so that it consumes maximum power? Determine the maximum power that can be delivered to this resistance.
3. Graph $P_{\mathrm{L}}/P_{\max}$ as a function of $\mathrm{R}_L$ Also graph the efficiency as a function of $R_{\mathrm{L}}$.

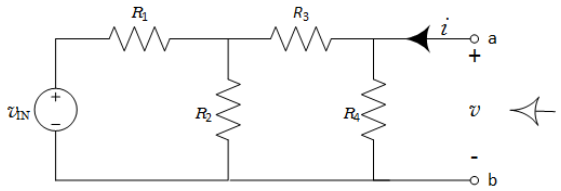



Figure 1: The Resistor Network for Pre-Lab Exercises

## 3 In-Lab Exercises

1. Construct the network shown in Figure 2.
   (a) Using a multimeter, measure the open circuit voltage and short circuit current.
   (b) Connect a 820 Ω resistance at the terminals *ab* and measure the voltage across it. Repeat the same process for resistance values: 1 kΩ, 1.2kΩ, 1.5 kΩ, 1.8kΩ, 2.2 kΩ, 2.7kΩ and 3.3 kΩ.

## 4 Post-Lab Exercises

1. From the measurements of In-Lab Exercise (1a), compute the Thevenin/Norton equivalent resistance of the network. How does your experiment compare with the pre-lab calculations? Comment on any discrepancies.
2. With reference to the measurements of In-Lab Exercise (1b)

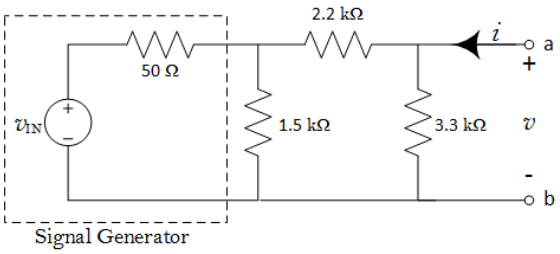



Figure 2: Resistor network for the In-Lab Exercises

(a) Compute the corresponding port current for each voltage-resistor pair.

(b) Using the calculations from (2a) above, graph the $i-v$ relation for the port. Also make an $i-v$ plot for the network using the measured open-circuit voltage and short-circuit current. Propose possible reasons for any deviations between the measured $i-v$ relation and the $i-v$ relation obtained from the Thevenin/Norton data. And also compare with the plots obtained during the pre-lab exercises..

3. Using the measurements of the In-Lab Exercise (1b)

(a) Calculate the power dissipated by each of the resistance values.

(b) Graph $P_{\mathrm{L}}/P_{\max}$ as a function of the load resistances. Also graph the efficiency as a function of load resistances. What is the efficiency of circuit at the point of maximum power transfer? Explain any discrepancies between this measured power data and the one found during the Pre-Lab Exercises.

---

⋉ ⋈ ⋊